MIT 8103 - Advanced Database Systems

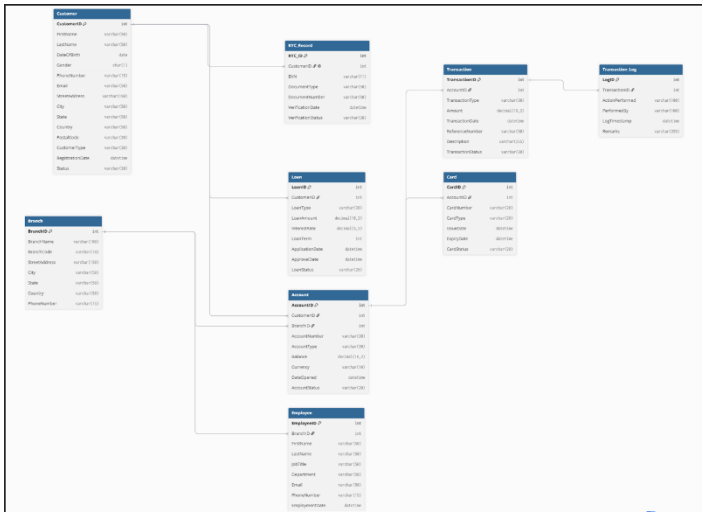
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Git Hub Repository URL: https://github.com/TosinKyle/MIT8103-Advanced-Database-Systems-Apex-Digital-Bank

Selected Organization - Apex Digital Bank - I selected Apex Digital Bank as the case study organization. Apex Digital Bank is a digital banking platform that provides financial services to individual and corporate customers through its web and mobile channels. The organization was used across all portfolios to demonstrate how database concepts, technologies and technologies apply in a realistic banking environment.

Portfolio 1: Database Design and Modelling

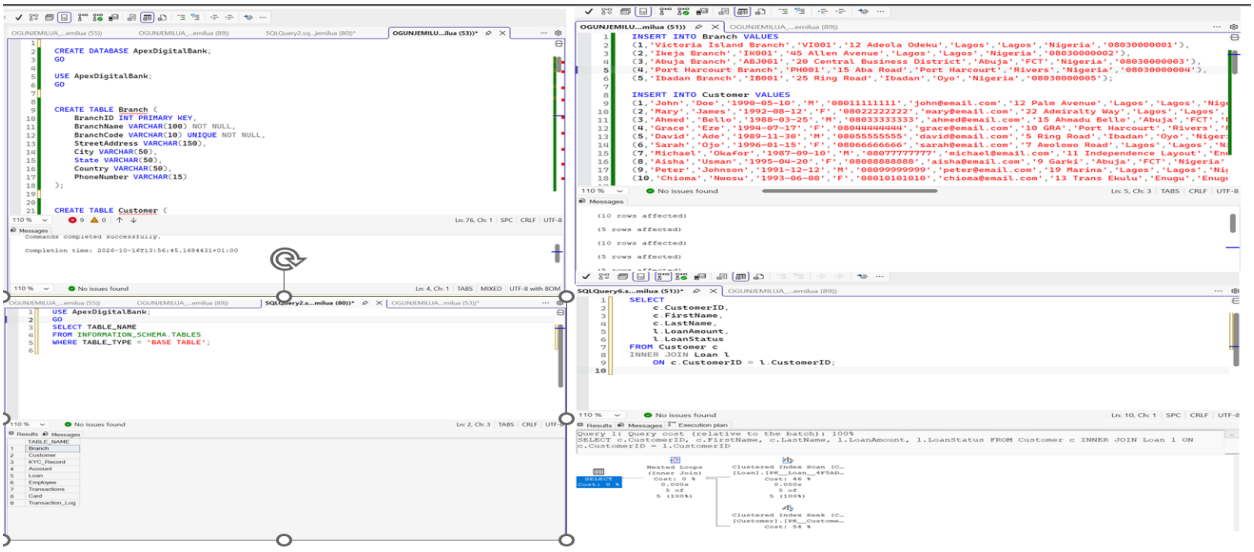
The first portfolio focused on designing a relational database for Apex Digital Bank. Business requirements were analyzed and key entities including Customers, Accounts, Transaction_Log, Transactions, Loans, Cards, Employees, Branches, and KYC records were identified.



An Entity Relationship Diagram was created to represent relationships between entities. A relational schema was then developed to ensure data consistency and accuracy. The database design was further refined through normalization to eliminate redundancy.

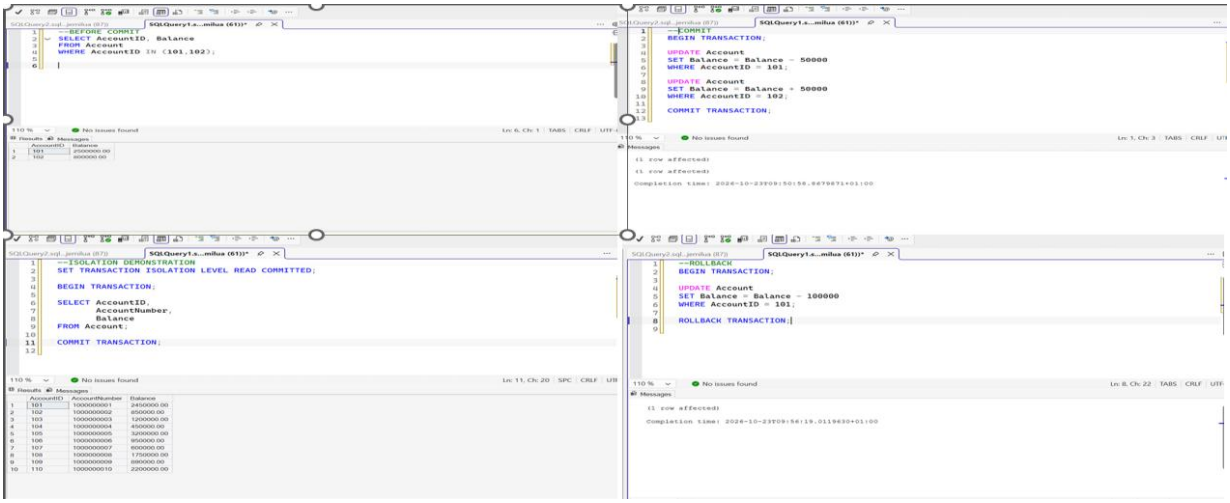
Portfolio 2: Query Processing and Optimisation

This portfolio concentrated on SQL query development and performance optimisation. Various SQL statements were developed to retrieve and manipulate banking data while query execution plans were reviewed to understand database processing behavior. Indexing and performance analysis were implemented to improve query performance and reduce execution time.



Portfolio 3: Transactions and Concurrency

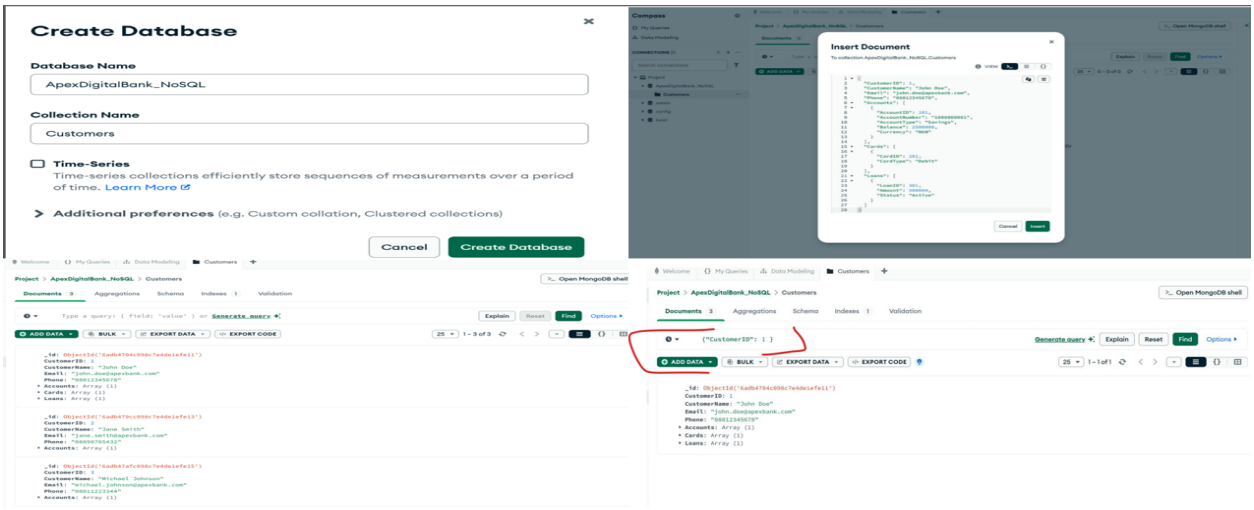
This portfolio explored transaction management and concurrency control in a multi-user database environment. Banking operations such as transfers were implemented using transactions to ensure data integrity. COMMIT and ROLLBACK operations were demonstrated to



show how successful and failed transactions are managed. Concurrency concepts and isolation mechanisms were examined to prevent issues such as inconsistent data states.

Portfolio 4: NoSQL and Advanced Data Models

This portfolio introduced NoSQL databases and advanced database concepts. MongoDB was used to implement a document oriented database. Additionally, NoSQL models were evaluated.



Portfolio 5: Distributed and Cloud Databases

The final portfolio examined distributed database systems and cloud database technologies. A distributed database architecture was proposed using primary and secondary database servers alongside a disaster recovery site.

The portfolio explored techniques including replication, sharding, availability, and consistency. It also evaluated distributed database principles such as ACID Properties, CAP Theorem, and the BASE Model. Finally, Microsoft Azure SQL Database was recommended as a cloud database platform capable of supporting the bank's requirements for scalability, high availability and secure data management.

Individual Reflection

By implementing this project, I had the opportunity to apply both theoretical and practical database concepts within a realistic business environment. I learnt how database systems support critical business operations and evolve to meet changing needs. Throughout the portfolios, I developed practical experience in relational database design, NoSQL databases, distributed and cloud database technologies. Overall, this module has strengthened my technical skills in database development, administration and optimization.

References & AI-Use Declaration

Microsoft Corporation. (2026). SQL Server Documentation &Azure SQL Database Documentation. https://learn.microsoft.com

MongoDB Inc. (2026). MongoDB Documentation. Available at: https://www.mongodb.com/docs

I used AI tools to support research, clarify concepts, organise content, and generate sample data for database demonstrations. All practical work was completed independently by me.